Thick Clouds Removing From Multitemporal Landsat Images Using Spatiotemporal Neural Networks

Yang Chen, Qihao Weng, *Fellow, IEEE*, Luliang Tang, Xia Zhang, Muhammad Bilal, and Qingquan Li

***Abstract*— Landsat images have played an important role in the field of Earth observation and geoinformatics. However, optical Landsat images are frequently contaminated by cloud cover, especially in tropical and subtropical regions, which limits the utilization of these images. To improve the utilization of Landsat images, in this study, we propose a novel spatiotemporal neural network with four modules: a cloud detection module, a spatial–temporal learning module, a spatial–temporal feature fusion module, and a reconstruction module. The results of the experiments demonstrate that the proposed method is quantitatively effective (root mean square error < 0.0179) and can achieve a better result for reconstructing Landsat images than some of the widely used existing deep learning methods and multitemporal methods. The proposed neural network method provides an effective tool for the removal of contiguous, thick clouds from satellite images, so as to improve the quality of subsequent remote sensing mapping and geoinformation extraction.**

***Index Terms*— Cloud detection, deep learning, geoinformatics, optical imagery, spatial–temporal feature fusion.**

I. Introduction

LANDSAT images have been applied in a wide range of applications, including resource surveys, information extraction, dynamic mapping, environmental and ecological assessments, urban growth analysis, and land-use and land-cover monitoring [1]–[3]. However, Landsat images are frequently contaminated by clouds and cloud shadows, especially in tropical and subtropical regions, which creates a large obstacle for the application of the Landsat data [4]. Therefore, it is an indispensable step to implement cloud removal for subsequent image analysis and applications.

A number of cloud-removal methods have been developed for optical satellite images. According to the source of supplementary data used, the existing methods can generally be classified into two types: single-temporal image-based methods and multitemporal image-based methods [5]. Single-temporal image-based methods rely on spectral- or frequency-domain information. For example, Zhang *et al.* [6] developed a haze optimization transformation method to remove the haze area from Landsat images. Xu *et al.* [7] used the cirrus band as auxiliary multispectral data to remove thin clouds. Xu *et al.* [8] proposed a model of noise-adjusted principal components transform (CR-NAPCT) to remove thin clouds from Landsat 8 data. Li *et al.* [9] used the two-stage haze and thin cloud removal method, based on homomorphic filtering (HF), for optical remote sensing data. However, as the thickness of clouds increases, the thick clouds can obstruct the majority and even all the land signals in the optical bands. Therefore, the abovementioned methods are not appropriate for thick cloud removal due to the lack of auxiliary spectral information.

Multitemporal image-based techniques have been widely applied to remove thick clouds. In these methods, multitemporal satellite images are utilized as auxiliary data to produce a cloud-free image. For example, Benabdelkader *et al.* [10] proposed a contextual reconstruction approach to remove thick clouds. Zhu *et al.* [11] developed a modified neighborhood similar pixel interpolator (MNSPI) method to remove thick clouds according to the relationship between a cloudy pixel and its neighboring pixels. Chen *et al.* [12] developed a novel spatially and temporally weighted regression (STWR) model to generate cloud-free Landsat images from multitemporal images. The cloud-contaminated regions are reconstructed by information cloning from multitemporal data [13]. Cheng *et al.* [14] proposed an effective method based on similar pixel replacement to generate cloud-free images from multitemporal high-resolution remote sensing images. The abovementioned methods of cloud removal used one auxiliary cloud-free image as reference data to fill the cloud-contaminated areas. Nevertheless, these methods directly assemble the spatiotemporal information from spatially or temporally similar region pixels.

Recently, deep learning methods have been successfully applied to many computer vision tasks for remote sensing imagery, such as image super-resolution [15], cloud detec-

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tion [16], and remote sensing image classification [17], [18]. Moreover, deep learning has been proven a more effective method for thin cloud removal in remote sensing images [19]–[21]. Qin *et al.* [19] proposed a novel multiscale CNN to remove thin clouds from optical remote sensing images. A residual symmetrical concatenation network (RSC-Net) was used to remove thin clouds from Landsat 8 images [20]. An attentive generative adversarial network (AGAN) was proposed by Qian *et al.* [21] to remove haze through a contextual autoencoder. The abovementioned deep learning methods are able to reconstruct the missing information for thin cloud areas. However, the performance of these methods is usually not satisfactory for thick cloud areas.

Multitemporal images should be considered as input auxiliary data for thick cloud removal [22]. Zhang *et al.* [23] developed a spatial–temporal–spectral (STS) feature-learning model to reconstruct cloud-contaminated areas. Chen *et al.* [24] proposed a content–texture–spectral convolution neural network (CTS-CNN) to reconstruct cloud-contaminated areas by extracting texture–temporal–spectral information from the multitemporal ZY-3 images. However, these methods require cloud-free images as the input auxiliary data. In other words, the performance of these methods depends on the availability of cloud-free images.

To address the limitations of these methods, synthetic aperture radar (SAR) images can also be considered as an auxiliary data source for cloud removal tasks. Eckardt *et al.* [25] introduced multifrequency SAR images to remove clouds from multispectral Landsat images. Generative adversarial networks (GANs) have been introduced to produce cloud-free images from SAR images [26]–[28]. Grohnfeldi *et al.* [26] developed conditional GANs to recover cloud-contaminated regions by exploiting SAR images. The cloud-GANs model developed by Singh and Komodakis [27] can further improve the reconstructed result in terms of spectral consistency. Bermudez *et al.* [28] exploited a novel conditional GAN to generate cloud-free images from SAR images. However, GANs may suffer from training and prediction instabilities when fed with input data with serious noise [29].

In summary, although the abovementioned methods have been widely applied to cloud removal, the existing methods have some common limitations.

1) Some of the existing methods can achieve satisfactory results, but they assume that cloud-free images as auxiliary data;
2) The performance of the existing methods is not satisfactory when cloud-contaminated images are not similar to the auxiliary cloud-free images;
3) Most of the existing methods are tested in small areas but cannot be extended for cloud removal of the whole Landsat image scene.

In this article, a novel spatiotemporal neural network (ST-net) is proposed to estimate the cloud-free image directly from multitemporal Landsat images. The main contributions of the proposed method are summarized as follows.

1) We propose a novel ST-net with four modules: a cloud detection module, a spatial–temporal learning module, a spatial–temporal feature fusion module, and a reconstruction module.
2) The designed spatial–temporal learning module employs multiscale convolutions to extract multilevel spatial–temporal information. The designed fusion module is used to adaptively combine spatiotemporal information learning module outputs to produce contiguous cloud-free Landsat images with a high spatial and spectral consistency and integrity.
3) Experiments with both simulated and real images were conducted to confirm the performance of the proposed method. The results of the experiments demonstrate that the proposed method can achieve a better result for reconstructing Landsat images than some of the widely used existing methods. The proposed neural network method provides a tool for the removal of contiguous, thick clouds from satellite images, so as to improve the quality of subsequent remote sensing mapping and geoinformation extraction.

The rest of this article is organized as follows. Section II introduces the proposed method. The experimental results are discussed in Section III. Section IV discusses the influence of the different network architectures for cloud removal results. The conclusion is summarized in Section V.

II. Methodology

Fig. 1 describes the proposed methodology of the present research, which consists of five steps: 1) input of the multitemporal Landsat images of the target area; 2) cloud detection; 3) extraction of spatiotemporal information; 4) fusion of spatiotemporal information with the cloud-covered areas using a novel spatiotemporal feature fusion module; and 5) retrieval of the actual information with high spatial and spectral accuracies for the cloud-covered area using the reconstruction module of the ST-net.

A. Cloud Detection Module

Accurate identification of cloud-contaminated pixels is a key step for cloud removal [30], [31]. The Function of mask (Fmask) 4.0 (https://github.com/gersl/fmask) can improve the cloud detection accuracy of Landsat images by reducing false-positive errors in cloud and cloud shadow detection using integrated the spectral information and contextual information [32]. Therefore, we use the Fmask 4.0 method to detect clouds and shadows. In the generated cloud and shadow masks, 0 represents the clear pixel, and 1 represents the cloud or cloud shadow pixel.

The Fmask 4.0 method consists of the following steps.

1) The inputs of the Fmask 4.0 method include Landsat top of atmosphere (TOA) reflectance, brightness temperature (BT) digital elevation models (DEMs), and global surface water occurrence data (https://global-surface-water.appspot.com).

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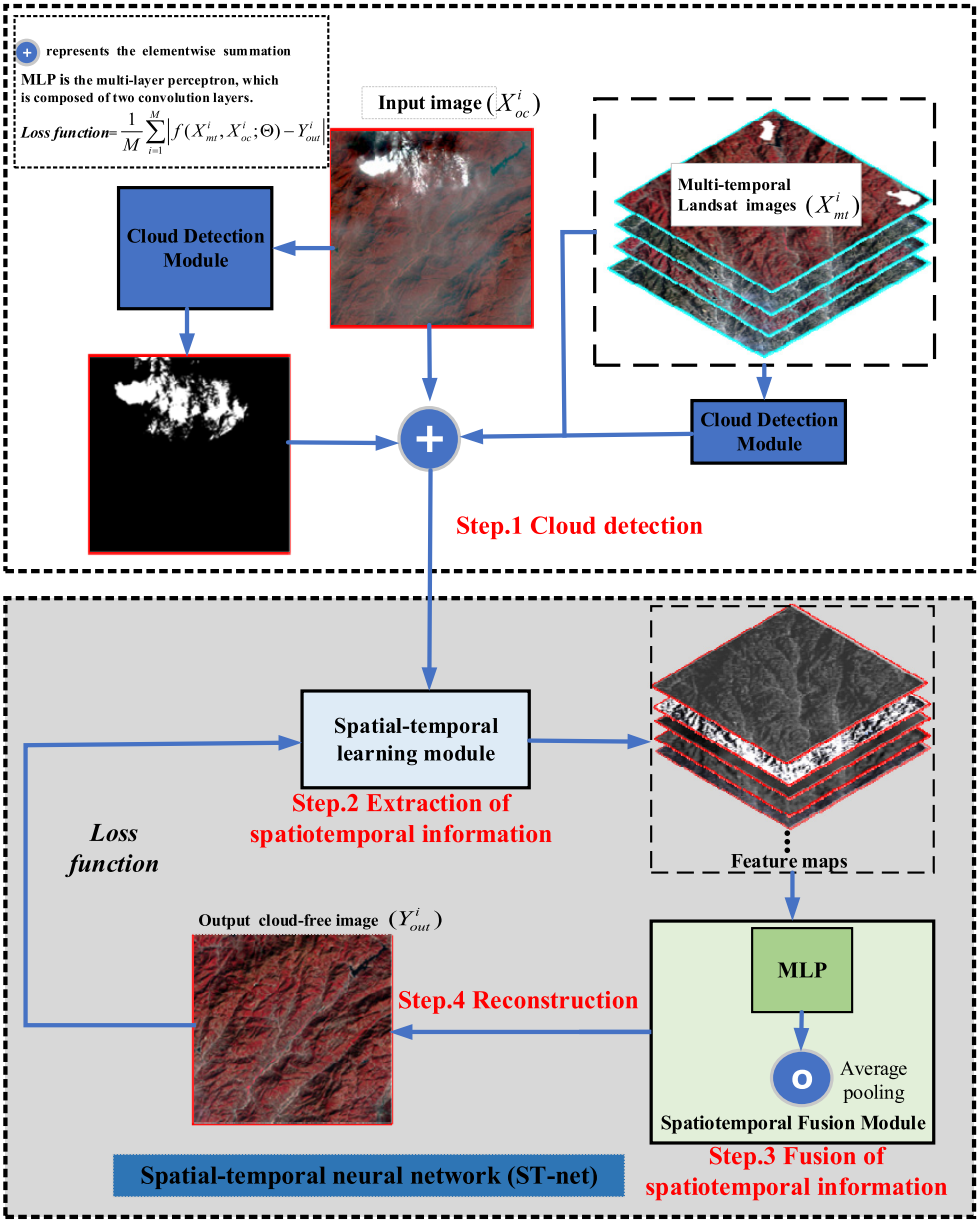


Fig. 1. Flowchart of the proposed clouds removal method.

2) The global surface water occurrence is the auxiliary data used to separate water and land pixels.
3) Brightness probability (Bright_W) and temperature probability (Temp_W) were calculated for water regions [31]. For water areas, the cloud probability is calculated by $\text{CP}_W = \text{Temp}_W \times \text{Bright}_W$.
4) Spectral variability probability (SVP_L) and temperature probability (Temp_L) were calculated for land [31]. For land areas, the cloud probability is calculated by $\text{CP}_L = \text{Temp}_L \times \text{SVP}_L$. In addition, spectral–contextual snow index (SCSI) was used to separate snow and cloud.
5) The clouds were detected by a dynamic threshold from cloud probability. The cloud shadows were detected by a geometry similarity match between clouds and their shadows.

B. Proposed ST-net Cloud Removal Architecture

It is not easy to extract multiscale spatiotemporal information when using traditional cloud removal methods [23]; however, the multiscale spatiotemporal information is important for automatic cloud removal. Based on the idea of image fusion [32], [33], we propose a novel cloud removal neural network for Landsat images, which is based on a convolution neural network framework to build a novel ST-net. The overall structure of the ST-net framework for cloud removal is shown in Fig. 2.

The ST-net cloud removal architecture is designed with a spatiotemporal information learning model, a spatiotemporal fusion module, and a reconstruction module. The ST-net cloud removal architecture is used to accomplish the spatiotemporal estimation from multitemporal images to target the

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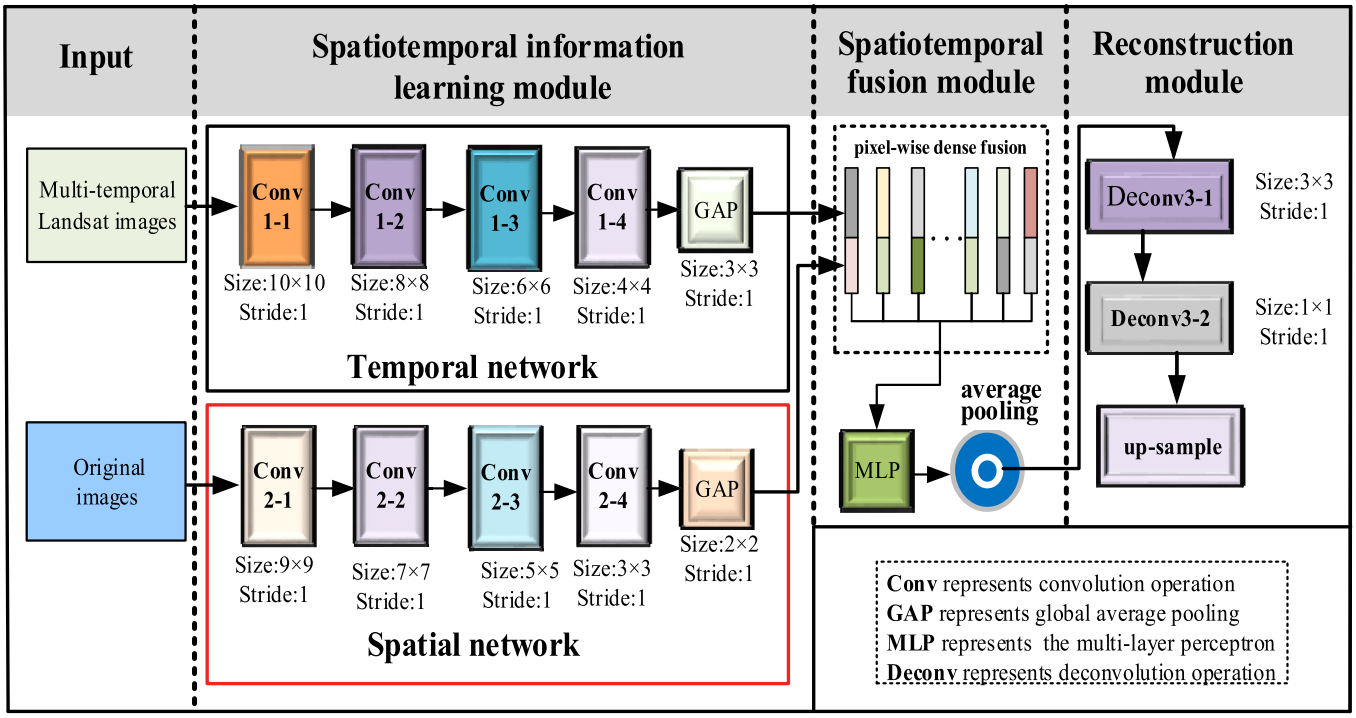


Fig. 2. Architecture of the proposed spatial–temporal neural network (ST-net) for cloud removal.

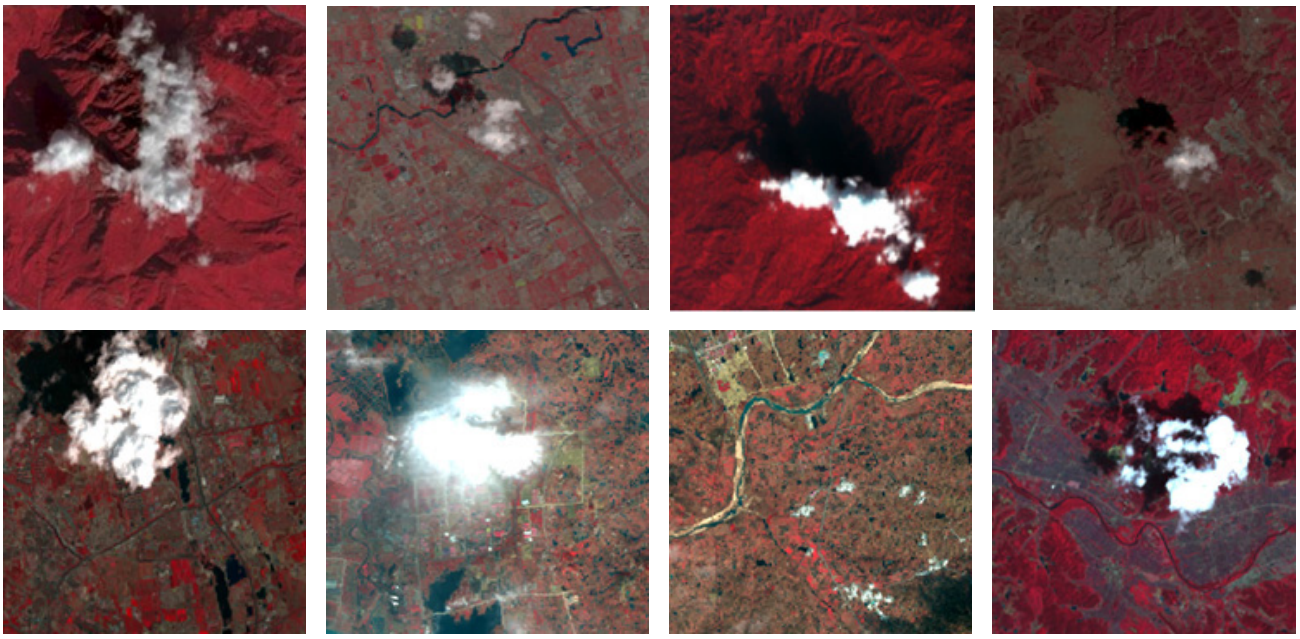


Fig. 3. Display of some of the training samples with different land-cover types.

cloud-free image. However, the highly linear spatial–temporal relationship between the cloud-contaminated image and auxiliary images indicates that robust modeling of the relationship and representation of spatial–temporal features are crucial factors for recovering cloud-contaminated pixels. In this study, the spatiotemporal information learning model is proposed to learn the highly nonlinear spatiotemporal relationship between cloud-contaminated images and auxiliary images, and the spatiotemporal fusion module aims to fuse the feature maps of the spatial information and temporal information. The details of each module are provided in Sections II-C and II-D.

Different from a common cloud removal network, the following holds.

1) The common cloud removal network always involves multiple layers of neurons, each of which extracts different level features, while, in the ST-net, the convolution kernels of different sizes are used to extract multilevel spatial–temporal information.

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2) In the ST-net, by replacing the fully connected layer with global average pooling, the number of learning parameters is heavily reduced, and the global spatiotemporal features are preserved.

3) The spatiotemporal dense fusion module is designed with the multilayer perceptron (MLP) and average pooling to fuse the feature maps of the spatial information and temporal information.

The ST-net method uses a time series of multiyear Landsat images. In this article, the multitemporal data preprocessing has main steps: 1) atmospheric correction; 2) cloud and cloud shadow masking using cloud detection module; 3) after obtaining the masks of multitemporal data, a sliding window procedure with a fixed stride is used to extract groups of corresponding cloud-free patches; and 4) the cloud-free patches are upsampled to the same size as the original multitemporal images.

The main advantages of the proposed ST-net cloud removal architecture are highlighted as follows: 1) different from traditional cloud removal methods, the input images of the proposed ST-net are multitemporal images of the same scene, which aims to get more complementary information; 2) the designed spatiotemporal information learning model (the spatial and temporal networks) employs multiscale convolutions to extract multiscale spatiotemporal information from the original spatial cloudy image and multitemporal images; and 3) the proposed ST-net cloud removal architecture can produce contiguous cloud-free Landsat images without using the cloud-free image as complementary data.

C. Spatiotemporal Information Learning Module

The spatiotemporal information learning model is based on two modified CNNs, consisting of a temporal network and a spatial network. The temporal network is designed to learn a high temporal correlation, while the spatial network extracts spatial information. It is not easy to reconstruct fine details when using high-level features [34]. The multiscale spatiotemporal content is a crucial factor for the precise reconstruction of cloudy regions [35]. Generally speaking, the smaller size convolutional kernels are used to exploit local features, and the larger size convolutional kernels are used to exploit global features [36], [37]. Therefore, we use the convolution kernel of different sizes to extract multilevel spatial–temporal information in the spatiotemporal information learning model.

We employed two subnetworks (spatial network and temporal network) to learn spatial and temporal features from the input original image and multitemporal images, respectively. The spatial convolutional block comprises four convolutional layers with sizes of 9, 7, 5, and 3. The temporal convolutional block comprises four convolutional layers with sizes of 10, 8, 6, and 4. These spatial and temporal networks have a similar framework but different convolution kernels. Each subnetwork contains four convolution layers, each followed by a rectified linear unit (ReLU). The spatial convolutional block is defined as follows:

$$G_S^k = f\left(\sum_{n=1}^{N}\left(W_S^{kn} \otimes G_{S-1}^n\right) + b^{(k)}\right) \tag{1}$$

where $b^{(k)}$ is the spatial convolutional bias vector, $f(\cdot)$ is the activation function, N is the number of spatial feature maps, and k represents the convolutional kernel sizes of 9, 7, 5, and 3. G_{S-1}^n and W_S^{kn} are the learnable parameters in the spatial convolutional kernels. The operation symbol $\otimes$ represents a convolution operation.

The temporal convolutional block is defined as follows:

$$G_T^m = f\left(\sum_{n=1}^{N}\left(W_T^{mn} \otimes G_{T-1}^m\right) + b^{(m)}\right) \tag{2}$$

where G_{T-1}^n and W_T^{mn} are the learnable parameters in the temporal convolutional kernels. $b^{(m)}$ is the temporal convolutional bias vector, and m represents the convolutional kernel sizes of 10, 8, 6, and 4.

D. Spatiotemporal Fusion and Reconstruction Modules

1) Spatiotemporal Fusion Module: The spatiotemporal information fusion module can generally be classified into two types: the spatiotemporal information concatenation fusion method and the spatiotemporal information elementwise fusion method [38]. The spatiotemporal information elementwise fusion method does not contain any parameter and can fuse a various number of feature maps. However, the spatiotemporal information elementwise fusion method cannot dense fusion spatiotemporal information from multiple feature maps. Therefore, spatiotemporal dense fusion module is designed with the MLP and average pooling, which can be mathematically expressed as (3). The MLP is a global average pooling layer that preserves the temporal–spatial features. The spatiotemporal fusion module aims to integrate the feature maps of the spatial information and temporal information with different convolutional kernel sizes. This module is used to fuse the spatial information and temporal information at the spatial position so that the band response at the same location is consistent

$$F(\mathbf{I}_{\text{out}}) = R(\text{MLP}(\text{avg}(\mathbf{I}_{\text{ins}}))) + R(\text{MLP}(\text{avg}(\mathbf{I}_{\text{int}}))) \tag{3}$$

where R is the ReLU activation function, $\mathbf{I}_{\text{ins}}$ denotes the feature maps generated by the global average pooling layer in the spatial network, $\mathbf{I}_{\text{int}}$ denotes the feature maps generated by the global average pooling layer in the temporal network, $F(\mathbf{I}_{\text{out}})$ is the final fusion feature map, and avg denotes the average pooling operation. In this article, to avoid parameter excessive, the MLP is composed of two convolution layers.

2) Reconstruction Module: The last stage of the proposed ST-net is recovering the desired cloud-free Landsat image from the fused features. Because the spatiotemporal information learning module only contains four convolutional layers, the abstraction level of the extracted convolutional information is not high.

We use deconvolution operation to reconstruct cloud-free Landsat image and upsample the feature map by a factor of 2 via bilinear interpolation to get output cloud-free image as the input image of the same size. The output expected cloud-free image is generated as follows:

$$Y_{\text{out}}^i = F(\mathbf{I}_{\text{out}}) + \mathbf{I}_{\text{int}} + V(\text{Concat}(\mathbf{I}_{\text{int}}, \mathbf{I}_{\text{ins}})) \tag{4}$$

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where Concat($\mathbf{I}_{\text{int}}, \mathbf{I}_{\text{ins}}$) are the whole input feature maps obtained by concatenating $\mathbf{I}_{\text{ins}}$ to $\mathbf{I}_{\text{int}}$ and $V(\text{Concat}(\mathbf{I}_{\text{int}}, \mathbf{I}_{\text{ins}}))$ is the feature maps produced by deconvolution kernel taking Concat($\mathbf{I}_{\text{int}}, \mathbf{I}_{\text{ins}}$) as input data.

The adversarial loss functions are often calculated as the mean square error (MSE) of the deep convolutional features of the reconstructed image and the cloud-free image [39]. However, the MSE loss could produce smooth reconstruction images due to the features of the L2-norm. Some studies have suggested that the L1-norm loss function is a good choice when performing remote sensing image fusion tasks [40]. In this study, the MSE L1-norm loss function in the ST-net is defined as

$$L_1(\Theta) = \frac{1}{M}\sum_{i=1}^{M}\left|f\left(\mathbf{X}_{mt}^{i}, \mathbf{X}_{oc}^{i}; \Theta\right) - \mathbf{Y}_{\text{out}}^{i}\right| \tag{5}$$

where $\mathbf{X}_{mt}^{i}$ and $\mathbf{X}_{oc}^{i}$ are multitemporal and the original images, respectively, and Θ is the function of network parameters. $\mathbf{Y}_{\text{out}}^{i}$ is the output expected cloud-free image. M is the number of training images. The f () function is the ReLU function.

The original cloudy image $\mathbf{X}_{oc}^{i}$, the original cloudy image mask $\mathbf{X}_{ocm}^{i}$, the multitemporal $\mathbf{X}_{mt}^{i}$, and the multitemporal mask $\mathbf{X}_{mtm}^{i}$ were integrally set as the input data. The workflow of the ST-net algorithm is shown as follows.

Algorithm 1 ST-net Algorithm for Cloud Removal

Input: $\mathbf{X}_{oc}^{i}$, $\mathbf{X}_{ocm}^{i}$, $\mathbf{X}_{mt}^{i}$, threshold L, and $\mathbf{X}_{mtm}^{i}$
Output: Cloud-free image $\mathbf{Y}_{out}^{i}$
Initialization: $i = 0$, $\mathbf{X}_{oc}^{0} = \mathbf{Y}_{out}^{i}$, $\mathbf{X}_{mt}^{0} = \mathbf{Y}_{out}^{i}$
Iteration: until $\mathbf{Y}_{out}^{i}$ is cloud-free image
for $i = 0, 1, 2, 3 \ldots$ to stopping rule **do**
  Extracting the spatiotemporal information by solving Eq. (1) and Eq. (2);
  Fusing the spatiotemporal information by solving Eq. (3);
  Reconstructing the current spatial patch in Eq. (4);
  If $L < 0.15$;
End Iteration
Output: The expected cloud-free image $\mathbf{Y}_{out}^{i}$

III. Experiment and Analysis

In this section, the performance of the designed ST-net is evaluated against both simulated and real images for multitemporal cloud-contaminated Landsat data. In the simulated experiments, different compositions of land covers were considered. In real data experiments, different types of cloud-contaminated Landsat images were tested. The proposed framework was evaluated against two traditional multitemporal methods and two state-of-the-art deep learning methods. The two traditional multitemporal methods include MNSPI [11] and STWR [12]. The MNSPI method can remove thick clouds according to the relationship between a cloudy pixel and its neighboring pixels. The STWR method makes full utilization of cloud-free information from input Landsat scenes to produce cloud-free Landsat images. The two state-of-the-art deep learning methods include CTS-CNN [24] and GANs [27]. The CTS-CNN is used to reconstruct cloud-contaminated areas by extracting texture–temporal–spectral information from the multitemporal ZY-3 images. The GANs are adopted to generate cloud-free images from SAR images. In this article, the CTS-CNN and the designed ST-net are trained with the same training data set. The GANs are trained with cloud-free samples from the Sentinel-1 data.

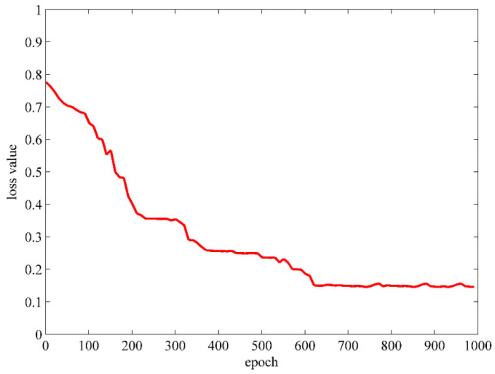


Fig. 4. Loss function of the proposed ST-net in the Landsat image data.

To quantitatively estimate the designed ST-net, the peak signal-to-noise ratio (PSNR) and root MSE (RMSE) are used [41]. The PSNR and RMSE are calculated as follows:

$$\text{PSNR} = 10\lg\frac{255^2}{\text{RMSE}} \tag{6}$$

$$\text{RMSE} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}[f(i,j) - g(i,j)]^2}{M \times N} \tag{7}$$

where $M \times N$ is the size of the image, (i, j) are the image coordinates, $G(i, j)$ is the gray value in the corresponding pixel, and $f(i, j)$ and $g(i, j)$ are the image pixel values before and after cloud removal, respectively.

A. Data Sets and Network Training

The experiment data were downloaded from the USGS Earth Explorer (https://earthexplorer.usgs.gov/). The experiment data were obtained from 20 pairs of Landsat data from different areas covering in Asia and Oceania. The experiment data are distributed in Asia (31°22′–39°26′ N, 115°05′–116°23′ E) and Oceania (49°36′–50°59′ S, 173°49′–175°49′ E). The Fast Line-of-Sight Atmospheric Analysis of Spectral Hypercubes (FLAASH) algorithm was used for atmospheric correction [42]. The proposed ST-net was trained and tested with samples from Landsat images with bands 1–7.

The proposed ST-net is trained and tested with samples from Landsat images. Fig. 3 shows some training samples. The acquisition dates of training and testing data range from 2003–2019 at the two sites (Asia and Oceania). The timing interval of multitemporal Landsat images is 16 days. The proposed ST-net was trained with cloud-contaminated Landsat images as the input data and multitemporal Landsat images as auxiliary data; 10 000 Landsat training samples were clipped from 15 pairs of Landsat images with a stride of 256 × 256 in ENVI software. These samples were used as the auxiliary image of the input cloud-contaminated Landsat testing samples

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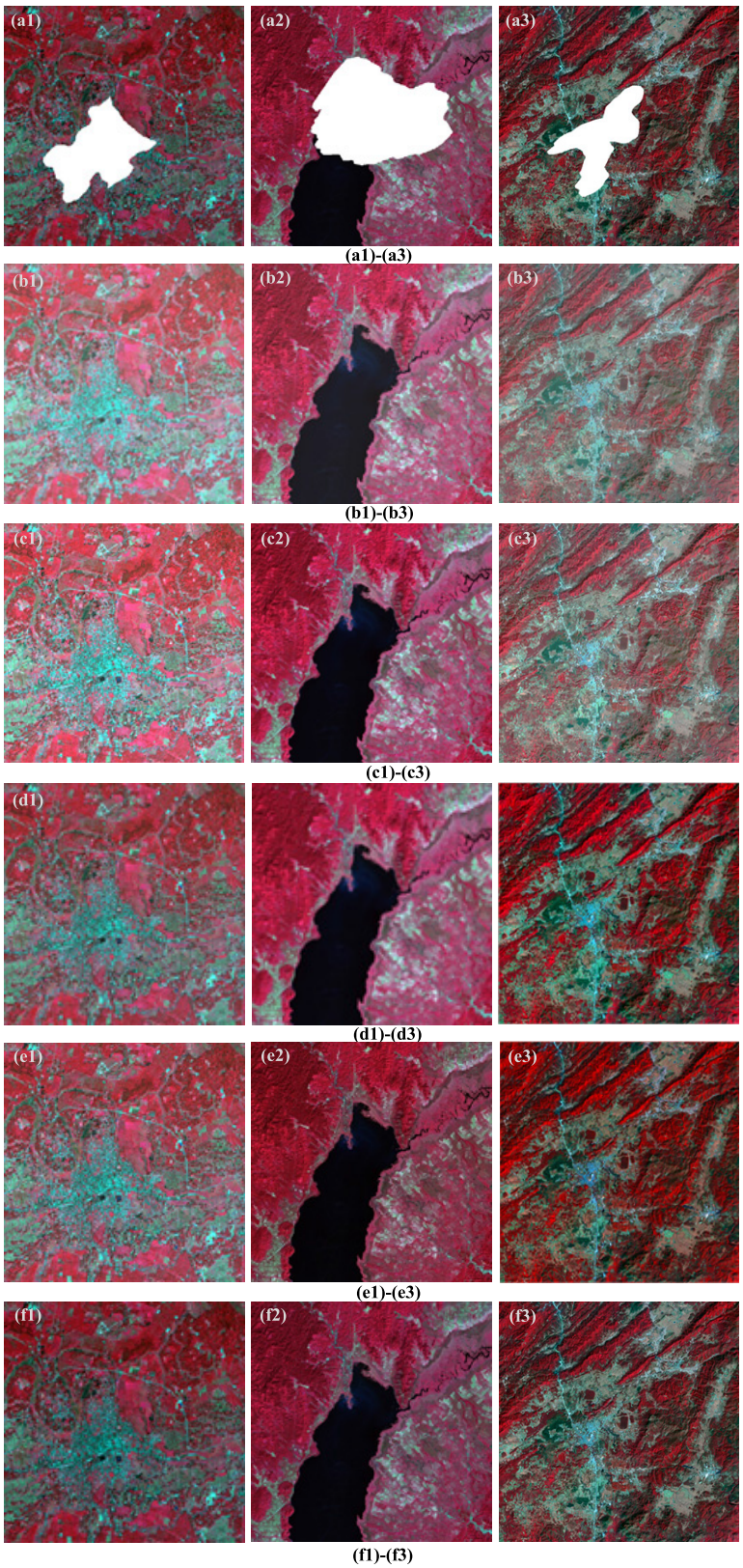


Fig. 5. Reconstruction results with the different methods for simulated Landsat 8 images (the false-color composite, 30-m spatial-resolution, from Beijing, China acquired on July 10, 2017, Path123/Row 32). (a1)–(a3) Simulated cloudy Landsat 8 images. (b1)–(b3) MNSPI method recover results. (c1)–(c3) STWR method recover results. (d1)–(d3) GANs method recover results. (e1)–(e3) CTS-CNN method recover results. (f1)–(f3) Proposed ST-net recover results.

was clipped from five pairs of Landsat images with a stride of 256 × 256 in ENVI software. For the testing data, the synthetic cloud masks were added into the original cloud-free data as the cloudy data for calculating training loss. It is worth noting that the input data in the training stage of the designed ST-net were only cloudy images.

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TABLE I
QUANTITATIVE EVALUATION OF THE RECOVER RESULT ON THE SIMULATED EXPERIMENT

Index	Bands	MNSPI [11]	STWR[12]	GANs[24]	CTS-CNN [21]	ST-net
RMSE (↓)	Band 1	0.0309	0.0278	0.0261	0.0254	0.0159
	Band 2	0.0294	0.0231	0.0221	0.0201	0.0138
	Band 3	0.0274	0.0204	0.0201	0.0198	0.0101
	Band 4	0.0195	0.0158	0.0144	0.0138	0.0056
	Band 5	0.0231	0.0161	0.0158	0.0141	0.0094
	Band 6	0.0243	0.0205	0.0188	0.0168	0.0142
	Band 7	0.0252	0.0226	0.0216	0.0190	0.0151
PSNR (↑)	Band 1	30.041	31.648	32.078	34.001	42.312
	Band 2	31.971	33.007	34.091	36.109	43.127
	Band 3	32.174	34.797	36.007	39.091	45.341
	Band 4	34.103	35.107	39.784	41.801	53.379
	Band 5	33.029	33.007	36.014	38.009	50.309
	Band 6	32.981	32.087	35.207	37.094	49.397
	Band 7	31.257	32.517	34.009	36.021	45.017

TABLE II
QUANTITATIVE EVALUATION OF THE RECONSTRUCTION RESULT FOR LANDSAT 5 IMAGE

Index	Bands	MNSPI [11]	STWR [12]	GANs [27]	CTS-CNN [24]	ST-net
Average RMSE (↓)	Band 1	0.0419	0.0386	0.0359	0.0318	0.0201
	Band 2	0.0378	0.0379	0.0299	0.0289	0.0168
	Band 3	0.0309	0.0337	0.0287	0.0271	0.0139
	Band 4	0.0286	0.0274	0.0259	0.0218	0.0079
	Band 5	0.0314	0.0296	0.0279	0.0241	0.0171
Average PSNR (↑)	Band 1	32.102	33.179	35.109	35.997	48.401
	Band 2	33.101	35.104	35.204	37.004	52.106
	Band 3	36.101	36.101	37.901	38.987	53.796
	Band 4	35.106	39.511	39.941	40.899	58.803
	Band 5	33.807	37.901	38.997	41.187	55.007

The experiments were conducted by using the software library TensorFlow. The designed ST-net is implemented using Python 3.5 over a personal computer with an E3-1505M v6@3 GHz, 32-GB DDR4 memory, and an NVIDIA Quadro M2200. The weights in each layer were initialized from a zero-mean Gaussian distribution with a standard deviation of 0.02. The learning rate started at 0.001 and was divided by 10 when the error plateaued. The weight decay and momentum were set to 0.001 and 0.1, respectively. We trained the proposed ST-net for 1000 epochs using stochastic gradient descent (SGD) [43] with a minibatch size of 256. Fig. 4 shows the loss functions during the training of the Landsat data after 1000 epochs.

B. Experiments on the Simulated Landsat 8 *Images*

In the simulation, different cloud contaminations are simulated to evaluate the performance of the proposed method under complex ground cover conditions (water, building, and vegetation). To analyze the influence of the different ground covers for the reconstruction, different cloud contaminations are simulated in the three Landsat 8 images [see Fig. 5(a1)–(a3)]. The masked area in Fig. 5(a1) shows an urban area, Fig. 5(a2) shows a rural area, and Fig. 5(a3) shows a mountainous region. Fig. 5(b1)–(b3) shows the corresponding cloud-free images.

To verify the performance of the designed ST-net, we compared the proposed method with the MNSPI method [11], the STWR method [12], CTS-CNN [24], and GANs [27]. Fig. 5 shows the reconstruction results for three different regions. The result of the proposed ST-net [see Fig. 5(f1)–(f3)] shows the most similarity with the original cloud-free images [see the images in Fig. 5(b1)–(b3)], which demonstrates the effectiveness of the method for maintaining the spatial and spectral integrity. The result of the MNSPI method [see Fig. 5(b1)–(b3)] shows that it cannot completely reconstruct the spectral information and texture information. Fig. 5(c1)–(c3) indicates that the STWR method failed to recover reasonable spectral details for different types of land covers. The GANs method [see Fig. 5(d1)–(d3)] showed a spectral distortion to some degree in complex background.

To quantitatively evaluate the reconstruction accuracy of the three different methods, the mean PSNR and mean RMSE values are computed for all bands (see Table I). Generally, lower RMSE and higher PSNR indicate better consistency of the reconstructed results with the original free-cloud images. Table I indicates that the proposed ST-net scored higher in PSNR and lower in RMSE than the other four methods. Thus, the simulated experiment shows that the proposed method is robust and can provide reasonably good reconstruction results for Landsat 8 images.

C. Experiments on Real Landsat 5 *Images*

To verify the efficacy of the designed ST-net, an experiment using real Landsat 5 images (bands 1–5) was implemented. The study area is located in Wuhan, China, and is covered by Path 123 and Row 39. This area contains water, buildings, and vegetation. To make a comparative analysis, the pro-

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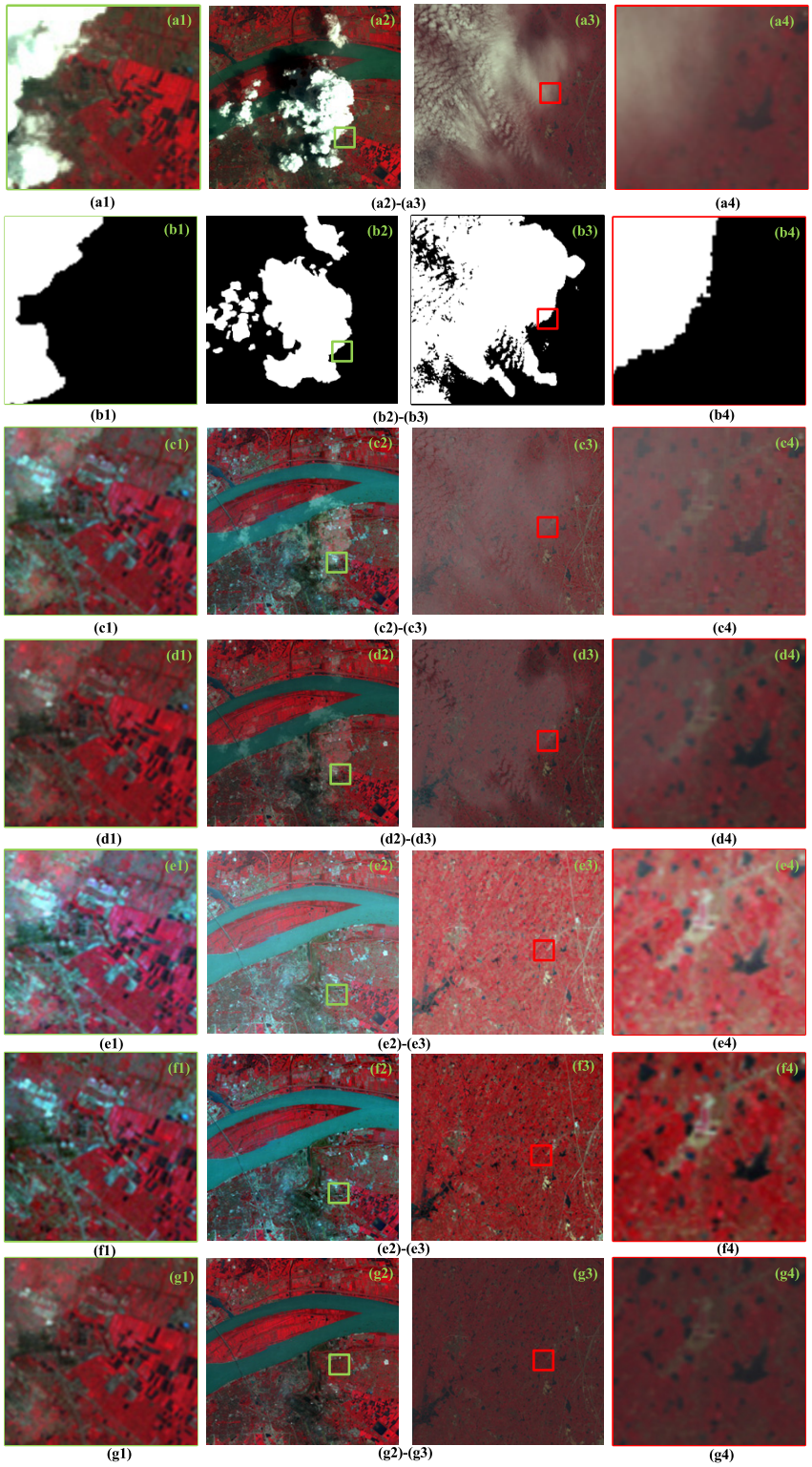


Fig. 6. Reconstruction results with the different methods for real Landsat 5 images (from Wuhan, China, acquired on August 5, 2009, at Path 123/Row 39). (a1) Local zoomed result of (a2). (a2) and (a3) Original Landsat 5 images. (a4) Local zoomed result of (a3). (b1) Local zoomed result of (b2). (b2) and (b3) Cloud and cloud shadow Mask of (a2) and (a3). (b4) Local zoomed result of (b3). (c1) Local zoomed result of (c2). (c2) and (c3) Result of the MNSPI method. (c4) Local zoomed result of (c3). (d1) Local zoomed result of (d2). (d2) and (d3) Result of the STWR method. (d4) Local zoomed result of (d3). (e1) Local zoomed result of (e2). (e2) and (e3) Result of the GANs method. (e4) Local zoomed result of (e3). (f1) Local zoomed result of (f2). (e2) and (e3) Result of the CTS-CNN method. (f4) Local zoomed result of (f3). (g1) Local zoomed result of (g2). (g2) and (g3) Result of the proposed ST-net method. (g4) Local zoomed result of (g3).

posed ST-net is compared with the MNSPI [11], STWR [12], CTS-CNN [24], and GANs methods [27].

Fig. 6 shows the reconstruction results by each method for the Landsat 5 image. The results of the two traditional

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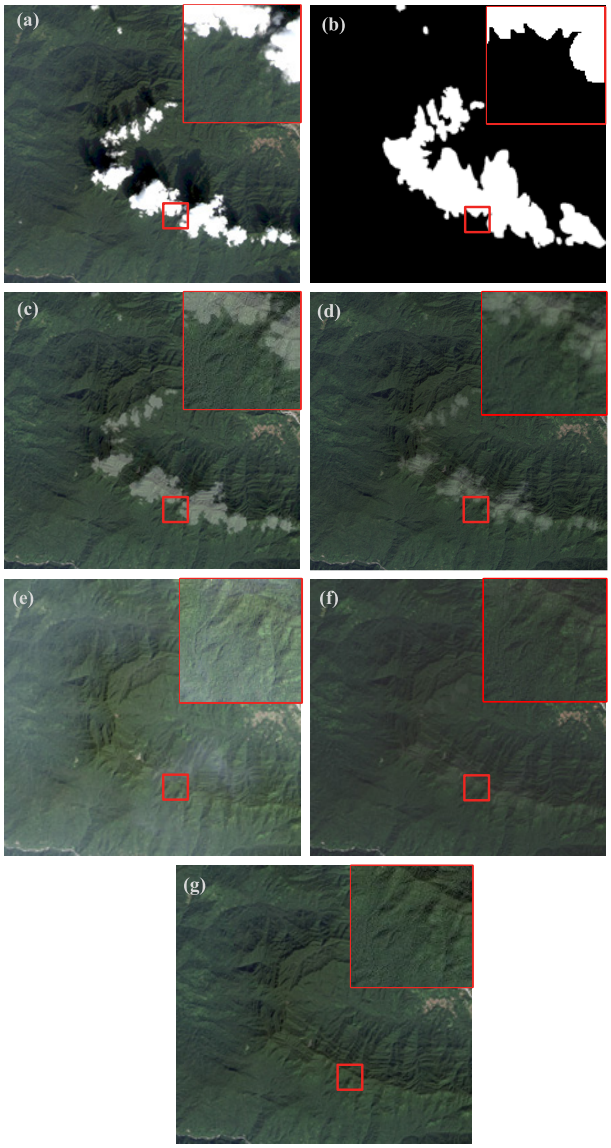


Fig. 7. Results of cloud removal with different methods for a real Landsat 8 image in true-color composite from Canberra, Australia, acquired on May 21, 2015, at Path 90/Row 84. (a) Original Landsat 8 image with clouds and shadows. (b) Cloud and cloud shadow Mask of (a). (c) Result of the MNSPI method. (d) Result of the STWR method. (e) Result of the GANs method. (f) Result of the CTS-CNN method. (g) Result of the proposed ST-net method.

multitemporal methods show that the MNSPI [see Fig. 6(c2) and (c3)] and the STWR method [see Fig. 6(d2) and (d3)] can effectively reconstruct some missing information, while other details are not reconstructed correctly, above both land and water. The result of the GANs method showed a spectral distortion to some degree and a loss of contrast. The CTS-CNN method performed reasonably well in recovering the missing information, but some structural details are blurred. The result of the proposed ST-net shows significant improvements in both reconstructing missing information and preserving the original spectral integrity.

The average RMSE and average PSNR values in Table II further show the efficiency of the proposed ST-net for cloud removal in Landsat 5 images compared with other methods.

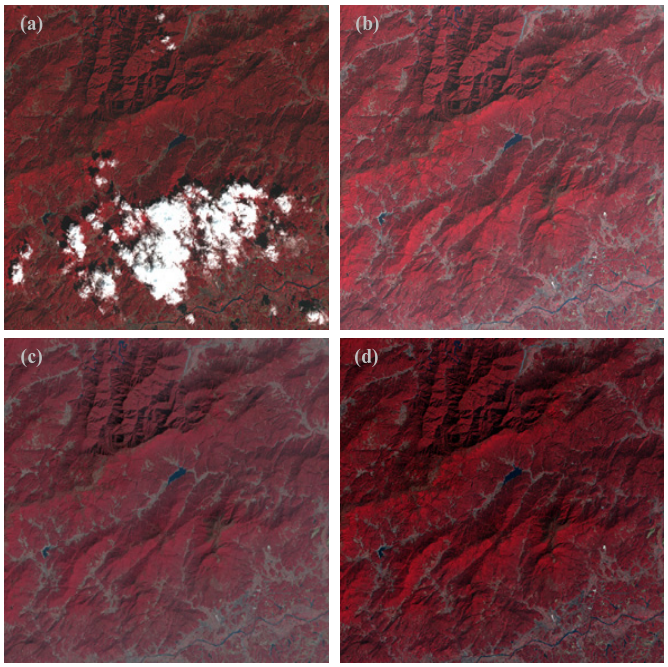


Fig. 8. Comparison of the different network models using real Landsat 8 image (from Wellington, New Zealand, acquired on August 5, 2016, at Path 73/Row 88). (a) Original Landsat 8 image. (b) Reconstruction result of the spatial network. (c) Reconstruction result of the temporal network. (d) Reconstruction result of the proposed ST-net method.

The proposed ST-net method obtained the largest PSNR and the smallest RMSE values for all reflected bands.

D. Evaluation of the Proposed Method on Landsat 8 Images

To further evaluate the performance of the designed ST-net, we compared the proposed method with the MNSPI [11], STWR [12], CTS-CNN [24], and GANs [27] methods for real Landsat 8 images with different types of clouds, cloud shadows, and haziness.

Fig. 7 shows the cloud removal results by using the MNSPI, STWR, CTS-CNN, GANs methods, and the proposed ST-net method. By comparing Fig. 7(g) with Fig. 7 (a), it can be seen that the proposed ST-net can remove different types of clouds, including thick, thin clouds, and cloud shadows. However, the MNSPI method [see Fig 7(c)] showed an obvious error in removing thick clouds, due particularly to the incorrect temporal information of the adjacent regions with significant changes. The results further showed that the STWR and the GANs methods [see Fig. 7(d) and (e)] can satisfactorily recover the surface structural features; nevertheless, the removal results showed spectral distortion for the entire image to some degree, which made the image appear hazy. The CTS-CNN method [see Fig. 7(f)] performed reasonably well in reconstructing cloud-contaminated areas; however, some texture details are missing. The RMSE and PSNR were calculated for the cloudy pixels before and after cloud removal to quantitatively evaluate the performance of the designed ST-net compared with its peers.

Table III shows significant errors in the MNSPI method in recovering cloud area information with low PSNR values. The

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TABLE III
QUANTITATIVE EVALUATION OF THE RECOVER RESULT ON A REAL LANDSAT 8 IMAGE

Index	Bands	MNSPI [11]	STWR[12]	GANs[27]	CTS-CNN [24]	ST-net
RMSE (↓)	Band 1	0.0369	0.0344	0.0311	0.0298	0.0179
	Band 2	0.0358	0.0312	0.0291	0.0281	0.0154
	Band 3	0.0301	0.0292	0.0287	0.0251	0.0112
	Band 4	0.0276	0.0261	0.0251	0.0198	0.0059
	Band 5	0.0284	0.0281	0.0264	0.0221	0.0091
	Band 6	0.0292	0.0304	0.0271	0.0235	0.0103
	Band 7	0.0327	0.0317	0.0297	0.0267	0.0149
PSNR (↑)	Band 1	32.372	33.014	34.678	36.608	49.341
	Band 2	34.107	34.678	35.994	37.009	50.974
	Band 3	34.931	35.697	37.012	38.797	52.196
	Band 4	38.694	39.471	40.108	42.907	60.003
	Band 5	37.131	38.104	39.017	39.107	58.009
	Band 6	35.364	36.048	38.127	37.497	56.081
	Band 7	34.301	34.601	35.031	36.917	54.057

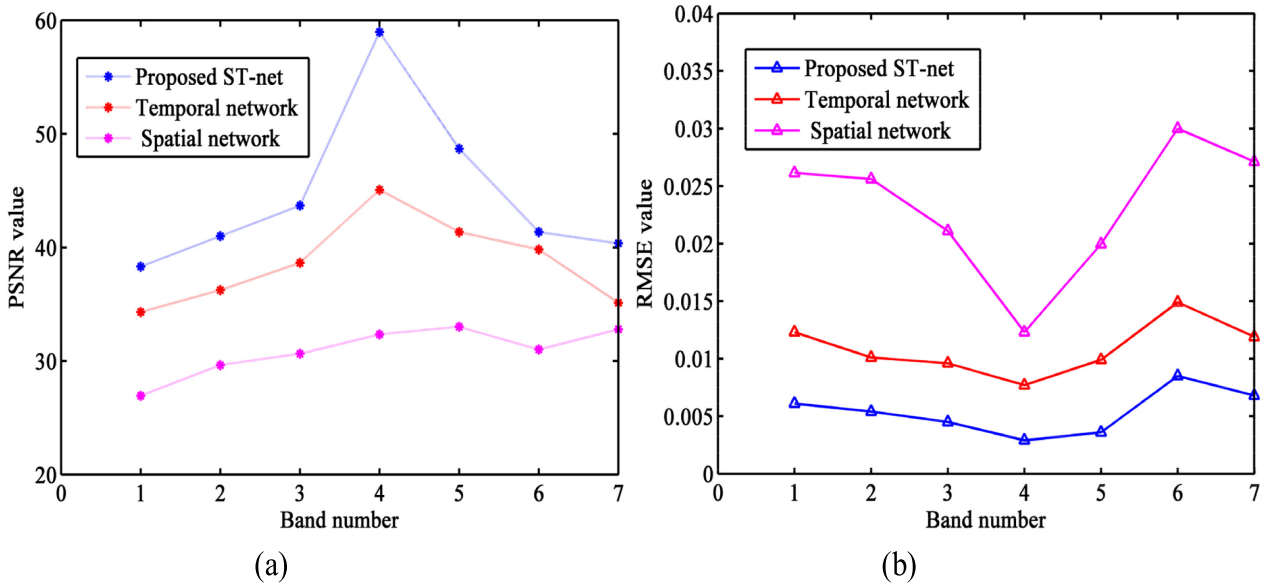


Fig. 9. Quantitative results of the different network architectures. (a) PSNR. (b) RMSE.

removal results of the STWR and the GANs methods show some spectral distortions by scoring a higher RMSE value for each band.

IV. DISCUSSION

In this section, the proposed network components are discussed for cloud removal. In the proposed ST-net architecture, the spatial and temporal networks are used to extract spatial and temporal features from original images and multitemporal images, respectively. Each of the networks uses multiscale convolution layers to extract multilevel spatial–temporal information. To verify the effects of the spatiotemporal fusion module, we compared the proposed ST-net with the spatial network (the proposed ST-net without fusion step and temporal network; see a black box in Fig. 2) and temporal network (the proposed ST-net without fusion step and the spatial network; red box in Fig. 2).

Fig. 8 shows the results of cloud removal with the proposed ST-net components for real Landsat 8 data. Fig. 8(b) and (c) shows that either the spatial or temporal network cannot recover cloud-contaminated regions accurately. In contrast, the reconstruction result of the proposed ST-net can accurately restore cloud-contaminated regions while preserving spectral integrity. The sensitivity of the different networks was further analyzed by computing the PSNR and RMSE for each band, as shown in Fig. 9. It can be seen that the proposed ST-net architecture achieved the best result in the terms of the RMSE and PSNR compared with the spatial and temporal networks. These results suggest that the proposed ST-net can provide a plausible reconstruction result for regions contaminated by clouds. In addition, the spatial network yielded the highest RMSE and lowest PSNR values. Overall, the spatial–temporal information can both provide key auxiliary information for cloud removal of optical images.

To further investigate the application range of the proposed ST-net, an entire Landsat 7 image is employed for the experiment, as shown in Fig. 10. The experimental region is located in Shandong, China, and is covered by Path 121 and Row

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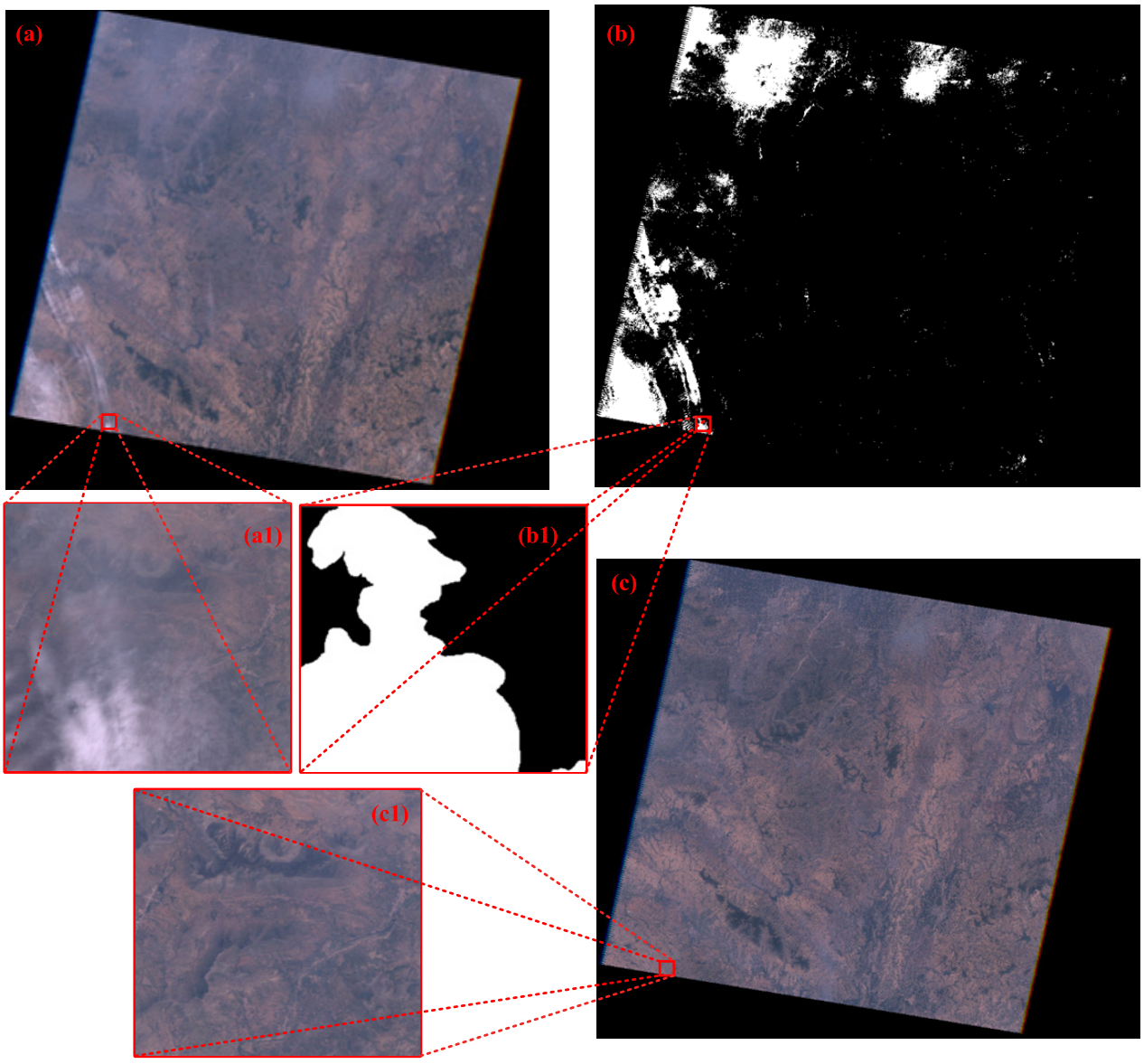


Fig. 10. Reconstructed entire Landsat 7 image using the proposed ST-net (30-m resolution; from Shandong, China, acquired on May 27, 2003, at Path 121/Row 35). (a) Original Landsat 7 image. (a1) Original image local zoomed. (b) Cloud and cloud shadow Mask of (a). (b1) Local zoomed result of (b). (c) Reconstruction image. (c1) Reconstruction image local zoomed.

35. Almost all cloud-contaminated pixels can be reconstructed by the proposed ST-net method. The proposed ST-net method yielded the largest PSNR (0.0121) and the smallest RMSE (50.304) values for all reflected bands in an entire Landsat 7 image. This experiment demonstrates that the proposed ST-net method is robust for reconstructing an entire Landsat 7 image scene.

V. Conclusion

In this article, we propose a novel ST-net to improve the utilization of Landsat satellite images. This method aims to solve a more complex and practical problem: contiguous, thick cloud removal. In the ST-net, the spatial–temporal learning module based on the convolution neural network is designed to extract multilevel spatial–temporal information from auxiliary multitemporal Landsat images. A novel spatial–temporal feature fusion module is then introduced to fuse the spatial information and temporal information. The reconstruction module is used to produce contiguous cloud-free Landsat images with high spatial and spectral consistencies and integrities. The results of the experiments in this study demonstrate that the proposed method can provide a better reconstruction result for Landsat compared with the existing multitemporal-based methods.

The proposed ST-net method can effectively and robustly remove thick clouds. However, the proposed ST-net also has some limitations. First, because the Fmask 4.0 method [30] cannot perform well for snow/ice and cloud detection, the proposed ST-net may not be efficient for bright land sites (snow, ice, and white roofs). Second, the proposed ST-net method fails to recover cloudy regions without auxiliary multitemporal data. In future work, we will continue improving the performance of the proposed ST-net by testing with image

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data, such as MODIS, Sentinel-2, and Sentinel-1, while trying to solve the problems identified earlier and by integrating multisource data for Landsat image reconstruction in order to build high-quality time-series image data sets for geoinformation extraction.

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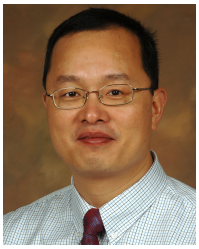
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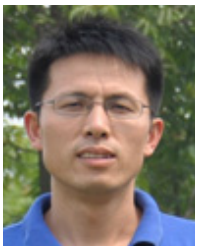
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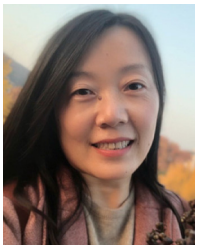
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